INCH - POUND

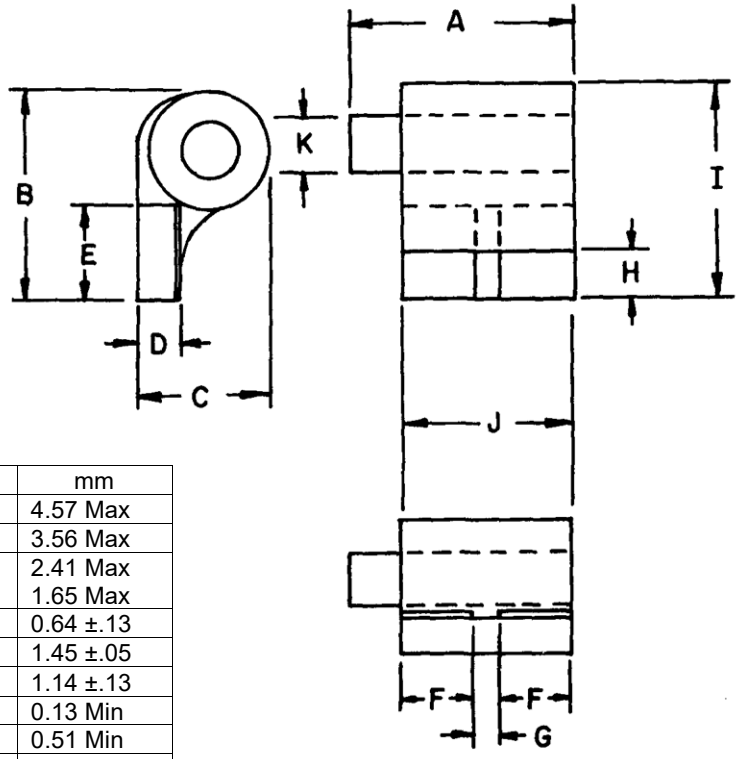
MIL-PRF-83446/12A
w/AMENDMENT 2
28 May 2024
SUPERSEDING
MIL-PRF-83446/12A
w/AMENDMENT 1
7 April 2019

PERFORMANCE SPECIFICATION SHEET

# COILS, RADIO FREQUENCY, CHIP, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-83446.




| Ltr | Inches | mm |
|---|---|---|
| A | .180 Max | 4.57 Max |
| B | .140 Max | 3.56 Max |
| C | .095 Max (-50 thru -61)<br>.065 Max (-1 thru -49) | 2.41 Max<br>1.65 Max |
| D | .025 ±.005 | 0.64 ±.13 |
| E | .057 ±.002 | 1.45 ±.05 |
| F | .045 ±.005 | 1.14 ±.13 |
| G | .005 Min | 0.13 Min |
| H | .020 Min | 0.51 Min |
| I | .140 Max | 3.56 Max |
| J | .102 ±.003 | 2.59 ±.08 |
| K | .033 ±.002 | 0.84 ±.05 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Fixed, chip coil.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5950

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Epoxy.

Weight: 0.5 gram, maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: MIL-STD-202-301, test voltage 700 volts rms.

Barometric pressure: MIL-STD-202-105, test condition C, (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Part or Identifying Number (PIN): The part number shall be in the following form.

| M83446/12- | 04 | B |
|---|---|---|
| | Sequentially assigned dash Numbers (see table I) | Termination finish (see MIL-PRF-83446) |



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TABLE I. Electrical characteristics (initial) and dash numbers.

| Dash number | Inductance ±10% (μH) | Q (min) | Q (typ) | Test frequency (MHz) 1/ | Self resonant frequency (min) MHz | DC resistance (max) ohms | Current (max) mA 2/ | Incremental current 3/ |
|---|---|---|---|---|---|---|---|---|
| 01 | 0.010 | 36 | 60 | 200 | 1200 | 0.020 | 750 | ---- |
| 02 | 0.012 | 30 | 60 | 200 | 1200 | 0.025 | 750 | ---- |
| 03 | 0.015 | 30 | 60 | 200 | 1200 | 0.035 | 750 | ---- |
| 04 | 0.018 | 30 | 60 | 200 | 1200 | 0.040 | 750 | ---- |
| 05 | 0.022 | 45 | 60 | 200 | 1000 | 0.045 | 750 | ---- |
| 06 | 0.027 | 30 | 60 | 200 | 1000 | 0.050 | 750 | ---- |
| 07 | 0.033 | 30 | 60 | 200 | 1000 | 0.055 | 640 | ---- |
| 08 | 0.039 | 30 | 55 | 200 | 1000 | 0.060 | 600 | ---- |
| 09 | 0.047 | 30 | 55 | 100 | 1000 | 0.065 | 550 | ---- |
| 10 | 0.056 | 30 | 55 | 100 | 1000 | 0.060 | 520 | ---- |
| 11 | 0.068 | 20 | 55 | 100 | 800 | 0.060 | 480 | ---- |
| 12 | 0.082 | 20 | 55 | 100 | 800 | 0.060 | 470 | ---- |
| 13 | 0.100 | 30 | 50 | 25.0 | 750 | 0.160 | 460 | ---- |
| 14 | 0.120 | 32 | 50 | 25.0 | 700 | 0.170 | 455 | ---- |
| 15 | 0.150 | 40 | 50 | 25.0 | 650 | 0.190 | 450 | ---- |
| 16 | 0.180 | 30 | 50 | 25.0 | 600 | 0.200 | 430 | ---- |
| 17 | 0.220 | 42 | 50 | 25.0 | 570 | 0.300 | 350 | ---- |
| 18 | 0.270 | 40 | 50 | 25.0 | 500 | 0.350 | 310 | ---- |
| 19 | 0.330 | 36 | 50 | 25.0 | 400 | 0.720 | 280 | ---- |
| 20 | 0.390 | 35 | 50 | 25.0 | 380 | 0.800 | 250 | ---- |
| 21 | 0.470 | 38 | 50 | 25.0 | 450 | 1.26 | 230 | ---- |
| 22 | 0.560 | 40 | 50 | 25.0 | 400 | 1.30 | 220 | ---- |
| 23 | 0.680 | 40 | 50 | 25.0 | 300 | 1.40 | 210 | ---- |
| 24 | 0.820 | 40 | 45 | 25.0 | 250 | 1.50 | 200 | ---- |
| 25 | 1.0 | 37 | 45 | 25.0 | 220 | 1.60 | 200 | ---- |
| 26 | 1.2 | 35 | 45 | 7.9 | 175 | 1.70 | 200 | ---- |
| 27 | 1.5 | 31 | 45 | 7.9 | 135 | 2.50 | 190 | ---- |
| 28 | 1.8 | 30 | 45 | 7.9 | 120 | 3.00 | 185 | ---- |
| 29 | 2.2 | 40 | 45 | 7.9 | 95 | 3.30 | 180 | ---- |

See footnotes at end of table.



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TABLE I. Electrical characteristics (initial) and dash numbers.

| Dash number | Inductance ±10% (μH) | Q (min) | Q (typ) | Test Frequency (MHz) 1/ | Self resonant frequency (min) MHz | DC resistance (max) ohms | Current (max) mA 2/ | Incremental Current mA 3/ |
|---|---|---|---|---|---|---|---|---|
| 30 | 2.7 | 32 | 45 | 7.90 | 85 | 3.65 | 175 | ---- |
| 31 | 3.3 | 32 | 50 | 7.90 | 70 | 4.00 | 170 | ---- |
| 32 | 3.9 | 32 | 50 | 7.90 | 60 | 2.10 | 165 | ---- |
| 33 | 4.7 | 32 | 45 | 7.90 | 50 | 2.30 | 165 | ---- |
| 34 | 5.6 | 35 | 45 | 7.90 | 40 | 2.30 | 165 | ---- |
| 35 | 6.8 | 43 | 50 | 7.90 | 30 | 1.30 | 165 | ---- |
| 36 | 8.2 | 40 | 45 | 7.90 | 22 | 4.00 | 120 | ---- |
| 37 | 10 | 40 | 45 | 7.90 | 20 | 2.90 | 105 | 58 |
| 38 | 12 | 35 | 45 | 2.50 | 15 | 3.50 | 91 | 53 |
| 39 | 15 | 35 | 40 | 2.50 | 13 | 2.30 | 87 | 50 |
| 40 | 18 | 35 | 45 | 2.50 | 12 | 3.70 | 81 | 46 |
| 41 | 22 | 35 | 45 | 2.50 | 10 | 4.00 | 77 | 37 |
| 42 | 27 | 38 | 45 | 2.50 | 9.0 | 5.00 | 73 | 34 |
| 43 | 33 | 35 | 45 | 2.50 | 9.0 | 5.70 | 69 | 32 |
| 44 | 39 | 35 | 45 | 2.50 | 6.5 | 6.50 | 65 | 29 |
| 45 | 47 | 35 | 45 | 2.50 | 6.2 | 6.00 | 63 | 29 |
| 46 | 56 | 35 | 40 | 2.50 | 6.0 | 6.30 | 60 | 28 |
| 47 | 68 | 30 | 45 | 2.50 | 5.8 | 6.70 | 57 | 26 |
| 48 | 82 | 35 | 45 | 2.50 | 5.2 | 7.50 | 55 | 23 |
| 49 | 100 | 33 | 40 | 2.50 | 4.0 | 13.0 | 50 | 20 |
| 50 | 120 | 33 | 40 | 0.790 | 4.0 | 14.0 | 40 | 20 |
| 51 | 150 | 35 | 40 | 0.790 | 3.9 | 16.0 | 36 | 20 |
| 52 | 180 | 35 | 40 | 0.790 | 3.5 | 18.0 | 33 | 19 |
| 53 | 220 | 35 | 40 | 0.790 | 3.2 | 21.0 | 28 | 18 |
| 54 | 270 | 35 | 40 | 0.790 | 3.1 | 23.0 | 25 | 17 |
| 55 | 330 | 35 | 40 | 0.790 | 2.8 | 37.0 | 20 | 16 |
| 56 | 390 | 35 | 40 | 0.790 | 2.5 | 39.0 | 17 | 16 |
| 57 | 470 | 35 | 40 | 0.790 | 2.3 | 45.0 | 15 | 15 |
| 58 | 560 | 35 | 40 | 0.790 | 2.2 | 47.0 | 13 | 14 |
| 59 | 680 | 35 | 40 | 0.790 | 2.1 | 62.0 | 12 | 13 |
| 60 | 820 | 35 | 40 | 0.790 | 2.1 | 78.0 | 11 | 11 |
| 61 | 1000 | 33 | 40 | 0.790 | 1.1 | 96.0 | 10 | 10 |

1/ Test frequency range 0.25 through 25 MHz: For electrical characteristics measurements, use the HP260-A Q Meter, HP4342A, HP250B RX Meter or equivalent with TF-301 test fixture, or equivalent (see figure 2). Fixture inductance (approximately 0.038 μH) and residual Q-meter inductance (approximately 0.01 μH) should be subtracted from indicated inductance.

Test frequency range 25 through 200 MHz: For electrical characteristics measurements, use the HP190-A Q Meter, HP260-A Q Meter, HP4342A, HP250B RX Meter or equivalent with TF-301 test fixture, or equivalent (see figure 2). Fixture inductance (approximately 0.009 μH) should be subtracted from indicated inductance.

2/ Maximum current allowed is not to exceed the specified temperature rise.

3/ Applicable to ferrite core only.



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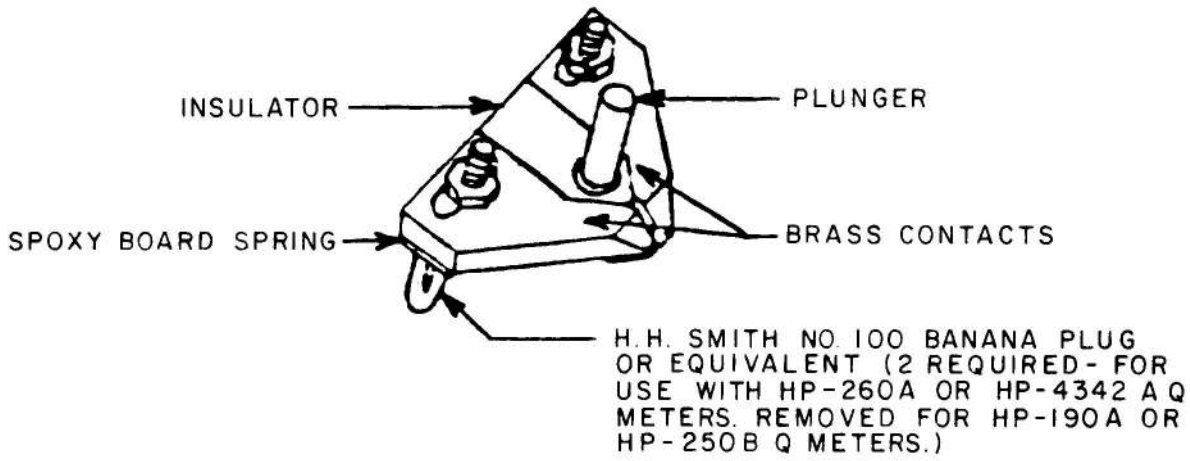



FIGURE 2. Typical chip coil test fixture (TF-301, or equivalent).

TABLE II. Electrical characteristics (final).

| Inspection group | Allowable variation from initial measurements | | | |
|---|---|---|---|---|
| | Inductance (Percent) | DC resistance | Self-resonant frequency (Percent) | Q (Percent) |
| Qualification inspection | | | | |
| Group II | ±5 | ±(3% +.001 ohm) | -8 | -10 |
| Group IV | ±5 | ±(2% +.001 ohm) | -10 | -10 |
| Group V | ±2 | --- | --- | -10 |
| Conformance inspection group C | | | | |
| Subgroup II | ±5 | ±(3% +.001 ohm) | -8 | -10 |
| Subgroup IV | ±5 | ±(3% +.001 ohm) | -8 | -10 |



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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-83446, this document references the following:

MIL-STD-202-105 MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA - CC

Review activities:
Army – MI
Navy – AS, OS
Air Force - 19

Preparing activity:
DLA – CC

(Project 5950-2024-010)

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